Requirements Gathering & System Specification Document

DATA ENGINEERING TRACK (AI & DATA SCIENCE) – ROUND 5

System Specification Charter

Project Title: Customer Churn Analysis & Retention Strategy Architecture
Scope Scale: 100,000 Subscriber Records × 100 Structural Features
Core Technology Stack: SQL ELT Pipelines (SQLite/PostgreSQL) → XGBoost/RF ML Pipeline → SHAP XAI → Streamlit App

1. OVERVIEW & REQUIREMENTS BASELINE

Requirements Gathering represents the foundational discovery phase necessary to define the operational, technical, and data specifications for the Telecom Customer Churn Prediction system. Operating within competitive telecommunications markets (e.g., Egypt's MNO market with 127.89M subscriptions and high MNP churn risk), this document codifies all data schemas, infrastructure constraints, model requirements, and business deployment objectives.

2. DATA & SCHEMA REQUIREMENTS (DATA INVENTORY)

The system ingests a multi-source relational schema totaling 100,000 instances and ~100 features. The core data inputs are defined as follows:

Data Source	Primary Fields & Schema	Key Features Included	Data Requirements & Joining Key
Client Demographics (`Client.csv`)	Demographic profiles (~51 structural features)	`customer_id`, `gender`, `plan_type`, `eqpdays` (equipment tenure)	Linked via unique primary key `customer_id`. Must handle missing categorical encodings.
Usage Logs (`Record.csv`)	Longitudinal behavior (~50 features)	`rev_Mean` (revenue), `mou_Mean` (minutes of use), `totcalls`, `churn`	Linked via foreign key `customer_id`. Contains target variable `churn` (binary 0/1).

3. TECHNICAL & DATA ENGINEERING REQUIREMENTS (ELT PIPELINE)

To process 100,000 subscriber instances efficiently without memory exhaustion, the following data engineering specifications are required:

Component	Technical Requirement	Specification / Implementation Details
Database Architecture	Relational SQL DB (SQLite / PostgreSQL)	Ingest raw CSVs into relational tables to perform scalable ELT joins.
Memory Management	Chunked SQL Processing	Process data in chunks (`chunksize=10000`) to prevent RAM overflow.
Preprocessing Engine	SQL Views & Automated Pipeline	Execute missing value imputation, standard scaling, and categorical encoding.
Class Imbalance	SMOTE Oversampling / Cost-Sensitive	Address ~15% baseline minority churn rate prior to model training.

4. MACHINE LEARNING & DEPLOYMENT REQUIREMENTS

- **Model Benchmark Specifications:** Train and evaluate Logistic Regression, Random Forest, and XGBoost models across an 80/20 train/test split.
- **Evaluation Metrics Priority:** Prioritize **Recall** and **ROC-AUC** over raw accuracy to minimize False Negatives (unidentified churners).
- **Explainable AI (XAI):** Integrate SHAP (SHapley Additive exPlanations) to explain individual subscriber risk drivers to customer service staff.
- **Interactive Web Deployment:** Deliver an operational **Streamlit / Flask Web Application** supporting real-time subscriber risk scoring and automated driver diagnostics.

5. PRESCRIPTIVE BUSINESS STRATEGY REQUIREMENTS

Predictive outputs must map directly to actionable retention campaigns: triggering subsidized handset upgrades for subscribers with extended device tenure (`eqpdays > 365`) and offering flexible billing tiers to price-sensitive customers.